

SOAP: Vision-Centric 3D Semantic Scene Completion with Scene-Adaptive Decoder and Occluded Region-Aware View Projection

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Abstract

Existing view transformations in vision-centric 3D Semantic Scene Completion (SSC) inevitably experience erroneous feature duplication in the reconstructed voxel space due to occlusions, leading to a dilution of informative contexts. Furthermore, semantic classes exhibit high variability in their appearance in real-world driving scenarios. To address these issues, we introduce a novel 3D SSC method, called SOAP, including two key components: an occluded region-aware view projection and a scene-adaptive decoder. The occluded region-aware view projection effectively converts 2D image features into voxel space, refining the duplicated features of occluded regions using information gathered from previous observations. The scene-adaptive decoder guides query embeddings to learn diverse driving environments based on a comprehensive semantic repository. Extensive experiments validate that the proposed SOAP significantly outperforms existing methods for the vision-centric 3D SSC on automated driving datasets, SemanticKITTI and SSCBench. Code is available at <https://github.com/gywns6287/SOAP>.

1. Introduction

Understanding 3D traffic scenes is crucial for autonomous driving systems to develop downstream tasks such as motion prediction and planning. LiDAR sensors [8, 33, 43, 54] provide precise 3D scene information, but their high cost and the complex setup limit their scalability. In contrast, areas of study about vision-centric 3D perception, including 3D object detection [11, 25, 46, 48, 52], 3D car shape reconstruction [14, 16, 17], and depth estimation [2, 27, 29, 45], have been receiving attention due to its simplicity and cost-effectiveness. However, these methods still struggle to re-

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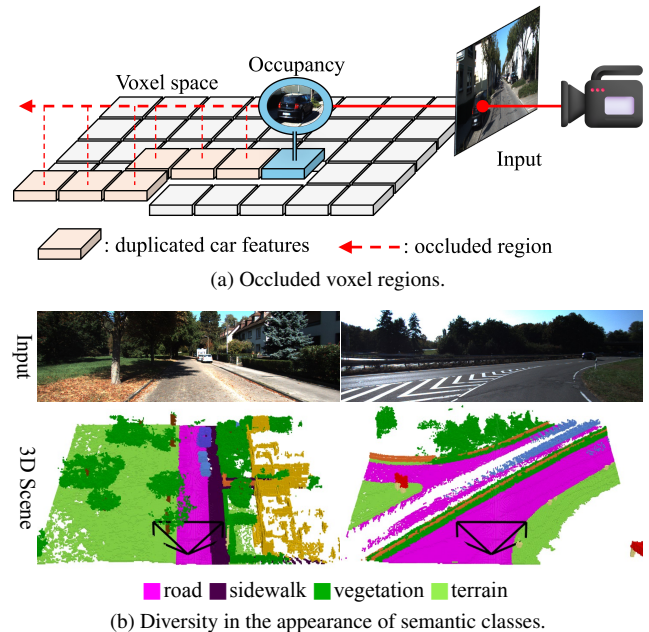


Figure 1. Two main motivations of SOAP. (a) The 2D-to-3D view projection inevitably yields erroneous feature duplication in voxel generation due to occlusion. (b) The semantic classes exhibit significant variability in their appearance depending on driving regions.

construct complete and fine-grained 3D scene representations. To address this limitation, 3D Semantic Scene Completion (SSC) has emerged as a promising alternative. 3D SSC leverages a voxel-based approach, assigning both an occupancy probability and a semantic label (e.g. car, pedestrian, road) to each voxel within the 3D space. Thereby, it can create a comprehensive representation of the driving scene. While earlier works [5, 19, 39, 41] primarily rely on LiDAR sensors, recent research [3, 13, 18, 22, 37, 50, 51] has demonstrated the potential of vision-centric methods.

A general vision-based 3D SSC model comprises an image encoder, a view projection module, and a voxel decoder. The image encoder first extracts 2D features from an input image, and then the view projection module subsequently lifts 2D image features into 3D voxel space. Given the 3D features of each voxel region, the voxel decoder estimates semantic classes corresponding to each voxel region. In this process, the precise 2D-3D view projection plays a crucial role. However, existing view projection methods [3, 50] inevitably experience erroneous feature duplication in the reconstructed voxel space due to occlusions, particularly along the longitudinal direction. For instance, as illustrated in Figure 1a, when a camera ray encounters an occupancy region (e.g. car), occluded regions (voxels behind this occupancy) may contain duplicated features of the obstructing car in the 3D view projection process. This duplication can occur even when the actual classes of the occluded regions are not related to a car, resulting in incorrect 3D SSC results.

Furthermore, the diverse appearances of semantic classes pose another significant challenge in 3D SSC. As shown in Figure 1b, semantic classes exhibit high variability in their appearance in real-world driving scenarios. For instance, road color and texture may vary depending on driving regions (e.g. urban or highway). Such appearance variations make it difficult for SSC models to distinguish similar classes, such as road $\leftrightarrow$ sidewalk, leading to potential misinterpretations of driving environments. While most previous works [3, 13, 22, 37] usually employ sequential 3D convolutions for the semantic decoder, this approach often lacks the flexibility to generalize for the diverse semantic context. More recently, Mask2former [6]-like approach [50, 51] is adopted for the SSC decoder, which utilizes semantic tokens derived from a query embedding to represent the semantic scene. However, these semantic tokens lack adaptability to dynamic driving environments, as they are fundamentally based on static query embedding.

In this work, we propose a novel 3D semantic scene completion network, SOAP, faithfully designed to address the occluded voxel regions and the visual variations in semantic classes. The proposed method leverages two key components: an occluded region-aware view projection and a scene-adaptive decoder. The occluded region-aware view projection (OAP) first converts 2D image features into 3D voxel features using the LSS algorithm [28], where these initial voxel features contain false duplicated information due to occlusions, as illustrated in Figure 1a. These occluded regions can be inferred by referring to previous frames, which provides valuable context about the missing observation. Therefore, OAP identifies occluded regions in the current frame using depth estimation and refines the duplicated features of occluded regions using information gathered from previous observations. Also, a vanilla transformer decoder [6] generates the semantic tokens based on

static query embedding, which is not versatile enough to capture a dynamic driving environment. To represent more diverse semantics, a repository builder first creates a semantic repository by gathering semantic tokens from various scenes. Then, the scene-adaptive decoder integrates informative clues from this extensive repository into the semantic tokens, guiding them to navigate diverse driving environments effectively.

Extensive experiments demonstrate that the proposed SOAP outperforms the other state-of-the-arts on two vision-centric SSC benchmarks, SemanticKITTI [1] and SSCBench-KITTI360 [21]. To summarize, the main contributions of our work are three folds:

- We develop the occluded region-aware view projection that effectively transforms 2D image features into 3D voxel space by recovering occluded voxel regions.
- We also introduce a scene-adaptive transformer that integrates informative semantic clues from the various scenes into the semantic tokens of an individual scene.
- The proposed SOAP achieves outstanding performances on two driving datasets, SemanticKITTI and SSCBench-KITTI360, with a margin of +2 and +2.67 mIoU scores, respectively.

2. Related Works

3D Semantic Scene Completion. 3D semantic scene completion (SSC) is an advanced task in 3D scene reconstruction fields, which aims to predict a dense semantic scene representation involving the geometrical occupancy and its corresponding semantic label. Early SSC methods [5, 19, 34] leverage 3D inputs to reconstruct voxelized indoor scenes. SSCNet [34] first defines the SSC task, which jointly reasons the geometry and semantics of an indoor scene from a single-depth image. 3DSketch [5] uses the Truncated Signed Distance Function to guide an occupancy and the semantic labels of the indoor scene. AICNet [19] develops anisotropic convolution to extract hybrid features from the depth and RGB images for indoor SSC tasks. Recently, the release of SemanticKITTI [1], the first large-scale outdoor SSC dataset, yields the development of diverse outdoor SSC methods. LMSCNet [31] uses the 2D UNet to extract collapsed BEV features from sparse 3D LiDAR scans and finally expands the height dimension for the 3D SSC. JS3C-Net [41] learns the contextual shape priors from sequential LiDAR data completing the sparse single sweep point cloud to the dense one for enhanced 3D SSC. The other existing works [7, 30, 36, 39, 40, 47] also commonly depend on 3D input such as LiDAR point clouds. In contrast, we address vision-centric 3D SSC, which only uses an RGB image for the semantic scene reconstruction.

Vision-centric 3D SSC. MonoScene [3] introduces the first vision-centric approach for SSC, utilizing a 3D UNet to process voxel features generated by sight projection. Vox-

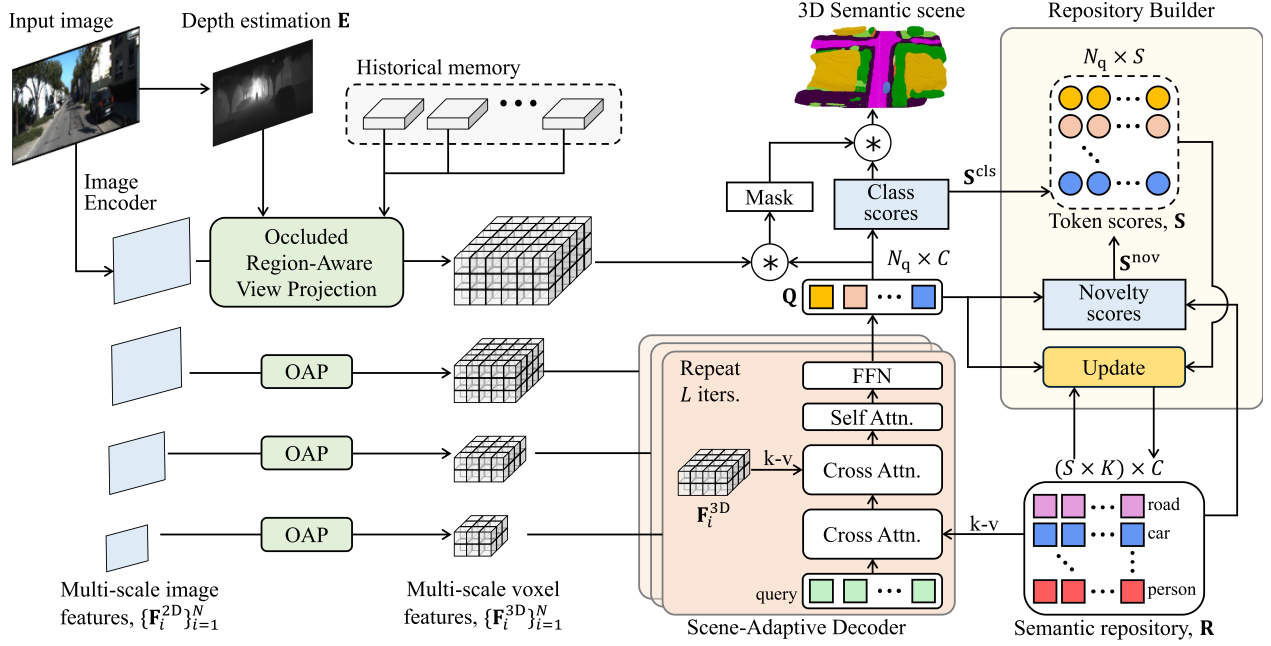


Figure 2. Overview of the proposed SOAP. Given multi-scale image features, an occluded region-aware view projection (OAP) transforms them into voxel space, refining duplicated features in occluded voxel regions using historical information. A scene-adaptive decoder enriches query embeddings with diverse semantic contexts through a semantic repository, which is constructed and iteratively updated by a repository builder.

Former [22] estimates occupancy regions using stereo depth estimation and selectively updates these regions with RGB images. HASSC [37] and Symphonies [13] also identify voxel occupancy based on stereo depth estimation, following VoxFormer. NDC-scene [42] extends a 2D feature map to a normalized device coordinates space to restore the depth dimension and subsequently reconstructs a 3D semantic scene using this space. SurroundOcc [38] adopts spatial 2D-3D attention to lift multi-scale 2D features into the 3D volume space. TPVFormer [12] introduces a query-based tri-perspective space for 3D SSC. Occformer [50] and LowRankOcc [51] reconstruct 3D scenes using semantic tokens on the Mask2former decoders [6]. SGN [26] propagates semantic contexts from semantic-aware seed voxels based on their spatial geometry cues. HTCL [18] leverages historical frames to refine current frame features through affinity-based dynamic refinement.

3. Method

3.1. Model Overview

3D Semantic Scene Completion (SSC) aims to predict semantic probabilities of the voxelized scene $\mathbf{V} \in \mathbb{R}^{X \times Y \times Z \times S}$, where (X, Y, Z) denotes the resolution of the 3D volume, and S is the number of semantic classes including a free (or empty) class. Figure 2 illustrates the overall architecture of the proposed SOAP. Image encoder first ex-

tracts multi-scale image features $\{\mathbf{F}_i^{2D} \in \mathbb{R}^{H_i \times W_i \times C}\}_{i=1}^N$ from an RGB image, where N is the number of scales, (H_i, W_i) is the spatial size at the scale i , and C is the channel dimension. Depth estimator predicts the depth map $\mathbf{E} \in \mathbb{R}^{H \times W}$ from RGB images. Consistent with previous approaches [13, 18, 22, 37, 44], we also use off-the shelf stereo depth model [32] for the depth estimator.

The occluded region-aware view projection (Section 3.2) utilizes the estimated depth map $\mathbf{E}$ to lift multi-scale image features $\{\mathbf{F}_i^{2D}\}_{i=1}^N$ into multi-scale voxel features $\{\mathbf{F}_i^{3D} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}\}_{i=1}^N$, enhancing occluded voxel regions based on informative features in historical frames. The scene-adaptive decoder (Section 3.3) first constructs a semantic repository using a repository builder and then it integrates semantic diversity from a semantic repository into query embeddings to produce scene-adaptive semantic tokens, which are utilized to reconstruct the final semantic voxel representation $\mathbf{V}$, in conjunction with multi-scale voxel features. Through this complementary process, SOAP guarantees a more flexible and accurate 3D SSC for a real-world driving environment.

3.2. Occluded Region-Aware View Projection

Occluded region-aware view projection (OAP) transforms 2D image features $\{\mathbf{F}_i^{2D}\}_{i=1}^N$ into 3D voxel space $\{\mathbf{F}_i^{3D}\}_{i=1}^N$. Figure 3 illustrates the detailed process of OAP. It produces voxel features for each i -th scale solely from the image feature at i -th scale.

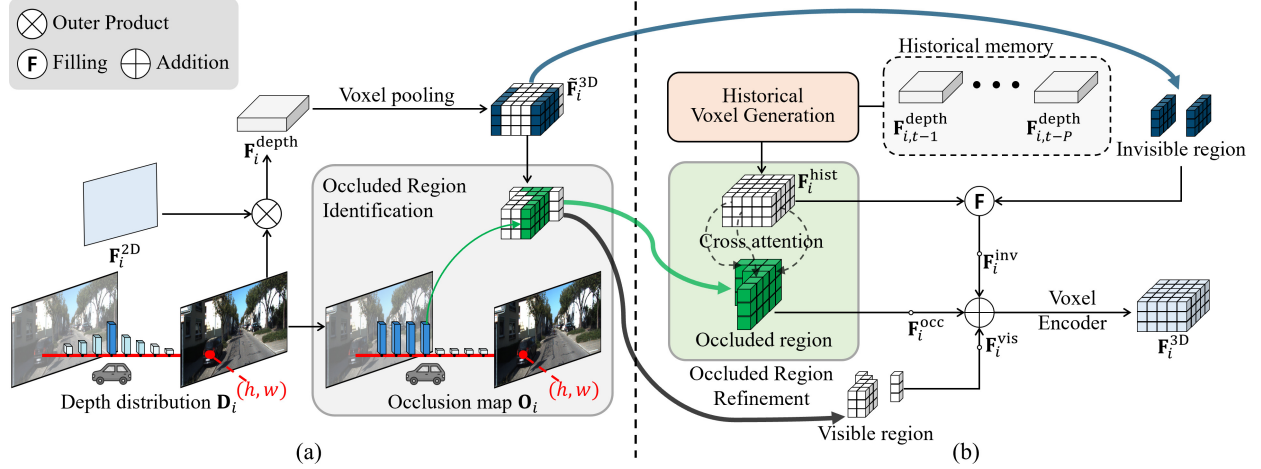


Figure 3. Ocluded region-aware view projection. (a) occluded region identification identifies occluded regions in the current frame by inferring an occlusion map based on the depth distribution. (b) occluded region refinement refines the duplicated features in the identified occluded regions using previous observations within the historical voxel features.

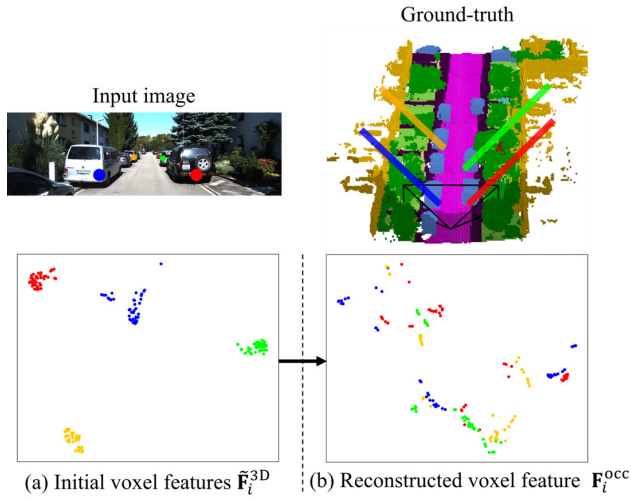


Figure 4. t-SNE visualization of voxel features in the occluded regions for (a) $\tilde{\mathbf{F}}_i^{3D}$ and (b) $\tilde{\mathbf{F}}_i^{occ}$. Different colors represent different camera rays.

Initial Voxel Generation. For each i th scale, OAP first generates initial voxel features $\tilde{\mathbf{F}}_i^{3D} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}$ from the 2D image features $\mathbf{F}_i^{2D}$ and the depth estimation $\mathbf{E}$. Specifically, OAP resizes the estimated depth $\mathbf{E}$ to $\mathbf{E}_i$ that has the spatial resolution of (H_i, W_i) , and distributes it to the depth distribution map $\mathbf{D}_i \in \mathbb{R}^{H_i \times W_i \times D}$. Here, the continuous depth value $\mathbf{E}_i(h, w)$ for each pixel (h, w) is converted into discrete depth bins $\mathbf{D}_i(h, w) \in \mathbb{R}^D$ represented as one-hot encoded vectors, followed by a Gaussian filter to create a soft depth distribution. Then, OAP computes the outer product $\mathbf{F}_i^{2D} \otimes \mathbf{D}_i$ to create the depth distributed features $\mathbf{F}_i^{\text{depth}} \in \mathbb{R}^{H_i \times W_i \times D \times C}$. The voxel-

pooling operation $\varphi(\cdot)$ in [28] can project $\mathbf{F}_i^{\text{depth}}$ into the voxel space using camera intrinsics and extrinsics. Thus, the initial voxel features are defined as $\tilde{\mathbf{F}}_i^{3D} = \varphi(\mathbf{F}_i^{\text{depth}})$.

Historical Voxel Generation. To represent previous observations, we incorporate depth distributed features $\{\mathbf{F}_{i,t-p}^{\text{depth}}\}_{p=1}^P$ from the previous frames into the current voxel space $\mathbf{F}_i^{\text{hist}} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}$, where P is the number of previous frames at timestamp t . Specifically, each past $\mathbf{F}_{i,t-p}^{\text{depth}}$ is projected into voxel space $\varphi(\mathbf{F}_{i,t-p}^{\text{depth}})$ and then aligned to the current voxel space at t via the warp-function $\mathcal{W}(\cdot)$ and summation operation:

$$\mathbf{F}_i^{\text{hist}} = \sum_{p=1}^P \mathcal{W}(\varphi(\mathbf{F}_{i,t-p}^{\text{depth}}), \mathcal{G}_t^{-1} \cdot \mathcal{G}_{t-p}) \quad (1)$$

where $\mathcal{G}_t$ and $\mathcal{G}_{t-p}$ are ego-pose matrices at t and $t-p$. These historical voxel features $\mathbf{F}_i^{\text{hist}}$ provide valuable perspectives that are not available in the initial voxel features.

Ocluded Region Identification. The initial voxel features $\tilde{\mathbf{F}}_i^{3D}$, which are projected from a single image view, inherently include false duplicated features due to occlusions. Figure 4a presents the t-SNE visualization of features in the occluded regions within $\tilde{\mathbf{F}}_i^{3D}$. Each point represents an individual voxel region, and voxel regions of the same color correspond to those located along the same camera ray. We observe that most occluded regions along the same camera ray exhibit similar patterns, which indicates erroneous replication of the features from the occupancy regions positioned in front of the occluded regions as in Figure 1a.

To address this limitation, OAP refines these occluded features by referring to previous frames, which provide valuable contexts about the missing observation in the current frame. As part of this process, we identify the occluded regions using the estimated depth distribution $\mathbf{D}_i$.

As in Figure 3a, each vector $\mathbf{D}_i(h, w)$ at the spatial location (h, w) contains the probability distribution of the nearest occupancy region along the longitudinal (or depth) direction. In other words, regions located behind the nearest occupancy with the highest depth probability often lose their semantic context information due to occlusions. To distinguish the occluded regions from non-occluded regions, we define an occlusion map $\mathbf{O}_i \in \{1, 0\}^{H_i \times W_i \times D}$ as

$$\mathbf{O}_i(h, w, d) = \begin{cases} 1 & \text{if } d > \arg \max_{\tilde{d} \in \{1, \dots, D\}} \mathbf{D}_i(h, w, \tilde{d}) \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

Figure 3a provides an example of $\mathbf{O}_i$, where occluded regions are marked with 1 (taller rectangular) and non-occluded regions are marked with 0 (shorter rectangular). To identify occluded regions in the voxel space, the occlusion map $\mathbf{O}_i$ is projected into voxel space through the voxel pooling operation, *i.e.* $\varphi(\mathbf{O}_i) \in \mathbb{R}^{X_i \times Y_i \times Z_i}$. Note that the regions identified by this occlusion map $\varphi(\mathbf{O}_i)$ contain incorrectly duplicated information from the nearest occupancy region along the longitudinal direction, as illustrated in Figure 4a. Therefore, we refine the features of these occluded regions through an occluded region refinement process.

Occluded Region Refinement. The occluded regions at the current time step may have been observable in previous time steps. Therefore, we refine deficient features in the occluded regions with previous observations in the historical voxel features $\mathbf{F}_i^{\text{hist}}$. To achieve this, we perform 3D deformable cross-attention, DCA [53] between the historical voxel features $\mathbf{F}_i^{\text{hist}}$ and the occluded voxel features, resulting in reconstructed occluded voxel features $\mathbf{F}_i^{\text{occ}} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}$:

$$\mathbf{F}_i^{\text{occ}}(x, y, z) = \begin{cases} \tilde{\mathbf{F}}_i^{3D}(x, y, z) + \text{DCA}(\tilde{\mathbf{F}}_i^{3D}(x, y, z), \mathbf{F}_i^{\text{hist}}), & \text{if } \varphi(\mathbf{O}_i)(x, y, z) \geq 1 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where (x, y, z) are spatial indices of the voxel space. This approach ensures that the reconstructed features $\mathbf{F}_i^{\text{occ}}$ effectively capture observable historical information about occluded regions, enabling more accurate and context-aware scene representation. As shown in Figure 4b, the reconstructed features $\mathbf{F}_i^{\text{occ}}$ include appropriately corrected information for the occluded regions, resulting in more diverse patterns according to different semantic regions.

Invisible and Visible Region Reconstruction. The initial voxel features also contain invisible regions due to the limited observation of a single camera. Since these invisible voxels do not contain any features, we directly fill them with the corresponding historical voxel features to recon-

struct invisible voxel features $\mathbf{F}_i^{\text{inv}} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}$:

$$\mathbf{F}_i^{\text{inv}}(x, y, z) = \begin{cases} \mathbf{F}_i^{\text{hist}}(x, y, z) & \text{if } \sum \tilde{\mathbf{F}}_i^{3D}(x, y, z) = 0 \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

We designate the voxel space, excluding invisible and occluded regions, as visible regions. The visible regions already contain appropriate semantic features with respect to the current observation, allowing us to directly retrieve the visible voxel features from the initial voxel features $\tilde{\mathbf{F}}_i^{3D}$. The visible voxel features $\mathbf{F}_i^{\text{vis}} \in \mathbb{R}^{X_i \times Y_i \times Z_i \times C}$ are defined as

$$\mathbf{F}_i^{\text{vis}}(x, y, z) = \begin{cases} \tilde{\mathbf{F}}_i^{3D}(x, y, z) & \text{if } \varphi(\mathbf{O}_i)(x, y, z) = 0 \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Voxel Encoder. To compute the final 3D voxel features, we sum the occluded, invisible, and visible voxel features, $\mathbf{F}_i^{\text{occ}} + \mathbf{F}_i^{\text{inv}} + \mathbf{F}_i^{\text{vis}}$, and process them through a voxel encoder to obtain the final voxel features $\mathbf{F}_i^{3D} \in \mathbb{R}^{X_i \times Y_i \times Z_i}$. In the voxel encoder, the spatial dimensions $(X_i \times Y_i \times Z_i)$ of the voxel features are sliced into Z_i bird's-eye view (BEV) planes with spatial resolution $(X_i \times Y_i)$. Then, two windowed self-attention blocks [23] are sequentially performed to each BEV plane.

3.3. Scene-Adaptive Decoder

A standard transformer decoder [6, 50, 51] yields semantic tokens starting from static query embeddings, which lack adaptability for dynamic driving scenes. To represent more diverse semantics, the scene-adaptive decoder correlates semantic tokens for a specific scene with tokens from different scenes. As in Figure 2, the scene-adaptive decoder generates scene-adaptive tokens $\mathbf{Q} \in \mathbb{R}^{N_q \times C}$ through L iterative transformer blocks, where N_q is the number of tokens. Each transformer block in the scene transformer cascades four modules including a cross-attention between query embeddings and the semantic repository, the other cross-attention between the query embeddings and the voxel features $\mathbf{F}_i^{3D}$, self-attention, and feed-forward networks. The cross-attention with the semantic repository integrates semantic features from dynamic scenes into the query embeddings, while the cross-attention with voxel features provides scene-specific semantic features. After the end of iterations, scene-adaptive tokens $\mathbf{Q}$ are projected to class predictions $\mathbf{S}^{\text{cls}} \in \mathbb{R}^{N_q \times S}$ and segmentation mask predictions $\mathbf{M} \in \mathbb{R}^{X \times Y \times Z \times N_q}$. Then, the final semantic voxel $\mathbf{V} \in \mathbb{R}^{X \times Y \times Z \times S}$ is formulated as

$$\mathbf{V} = \mathcal{F}^{-1}(\mathcal{F}(\mathbf{M})\mathbf{S}^{\text{cls}}) \quad (6)$$

where $\mathcal{F} : \mathbb{R}^{X \times Y \times Z \times C} \rightarrow \mathbb{R}^{XY \times Z \times C}$ is as spatial flatten and $\mathcal{F}^{-1}$ is its inverse operation.

Table 1. Semantic scene completion results on SemanticKITTI test set. † represents temporal methods that leverage previous frames for 3D SSC. We highlight the best results in **bold** and the second best results in underline.

Method	SC IoU	SSC mIoU	road (15.30%)	sidewalk (11.13%)	parking (1.12%)	other-ground (0.56%)	building (14.1%)	car (3.92%)	truck (0.16%)	bicycle (0.03%)	motorcycle (0.03%)	other-veh. (0.20%)	vegetation (39.3%)	trunk (0.51%)	terrain (9.17%)	person (0.07%)	bicyclist (0.07%)	motorcyclist (0.05%)	fence (3.90%)	pole (0.29%)	traf.-sign (0.08%)
MonoScene [3]	34.16	11.08	54.7	27.1	24.8	5.7	14.4	18.8	3.3	0.5	0.7	4.4	14.9	2.4	19.5	1.0	1.4	0.4	11.1	3.3	2.1
TPVFormer [12]	34.25	11.26	55.1	27.2	27.4	6.5	14.8	19.2	3.7	1.0	0.5	2.3	13.9	2.6	20.4	1.1	2.4	0.3	11.0	2.9	1.5
SurroundOcc [38]	34.72	11.86	56.9	28.3	30.2	6.8	15.2	20.6	1.4	1.6	1.2	4.4	14.9	3.4	19.3	1.4	2.0	0.1	11.3	3.9	2.4
VoxFormer† [22]	43.21	13.41	54.1	26.9	25.1	7.3	23.5	21.7	3.6	1.9	1.6	4.1	24.4	8.1	24.2	1.6	1.1	0.0	13.1	6.6	5.7
OccFormer [50]	34.53	12.32	55.9	30.3	31.5	6.5	15.7	21.6	1.2	1.5	1.7	3.2	16.8	3.9	21.3	2.2	1.1	0.2	11.9	3.8	3.7
NDC-Scene [42]	33.87	11.55	56.2	28.7	28.0	5.6	15.8	19.7	1.8	1.1	1.1	4.9	14.3	2.6	20.6	0.7	1.7	0.4	11.2	3.2	1.7
LowRankOcc [51]	38.47	13.56	52.8	27.2	25.1	8.8	22.1	20.9	2.9	3.3	<u>2.7</u>	4.4	22.9	8.9	20.8	2.4	1.7	2.3	14.4	7.0	7.0
Symphonies [13]	42.19	15.04	58.4	29.3	26.9	11.7	24.7	23.6	3.2	<u>3.6</u>	2.6	<u>5.6</u>	24.2	10.0	23.1	<u>3.2</u>	1.9	<u>2.0</u>	16.1	7.7	8.0
HASSC† [37]	42.87	14.38	55.3	29.6	25.9	11.3	23.1	23.0	2.9	1.9	1.5	4.9	24.8	9.8	26.5	1.4	<u>3.0</u>	0.0	14.3	7.0	7.1
SGN† [26]	<u>45.42</u>	15.76	60.4	31.4	28.9	8.7	<u>28.4</u>	25.4	<u>4.5</u>	0.9	1.6	3.7	<u>27.4</u>	<u>12.6</u>	28.4	0.5	0.3	0.1	18.1	<u>10.0</u>	8.3
HTCL† [18]	44.23	<u>17.09</u>	64.4	34.8	33.8	<u>12.4</u>	25.9	<u>27.3</u>	5.7	1.8	2.2	5.4	<u>25.3</u>	<u>10.8</u>	<u>31.2</u>	1.1	3.1	0.9	<u>21.1</u>	9.0	<u>8.3</u>
SOAP† (proposed)	46.09	19.09	<u>63.6</u>	36.2	36.8	17.2	28.7	28.9	3.3	5.0	5.3	7.0	29.8	15.1	32.3	3.7	2.3	0.2	22.5	11.7	13.1

Repository Builder. The semantic repository $\mathbf{R} \in \mathbb{R}^{(S \times K) \times C}$ stores K representative tokens for each semantic class, iteratively updated by semantic tokens with top- K highest scores. The detailed process of the repository builder is shown in Figure 2. Even though representative tokens could be selected using the classification scores $\mathbf{S}^{\text{cls}}$ of the scene-adaptive tokens $\mathbf{Q}$, relying solely on $\mathbf{S}^{\text{cls}}$ makes a repository filled with similar token and consequently degrades semantic diversity. To retain the semantic diversity within the repository $\mathbf{R}$, we introduce a novelty-based token selection. Specifically, we re-define token scores $\mathbf{S} \in \mathbb{R}^{N_q \times S}$ as $\mathbf{S} = \mathbf{S}^{\text{cls}} + \alpha \mathbf{S}^{\text{nov}}$, where $\mathbf{S}^{\text{nov}} \in \mathbb{R}^{N_q \times S}$ is novelty scores and α is its balancing parameter. To compute the novelty scores, the K tokens within each class in $\mathbf{R}$ are first averaged, resulting in the class average repository $\bar{\mathbf{R}} \in \mathbb{R}^{S \times C}$. Then, the novelty scores $\mathbf{S}^{\text{nov}}$ are calculated based on cosine similarity between each token in $\mathbf{Q}$ and $\bar{\mathbf{R}}$:

$$\mathbf{S}^{\text{nov}}(n, s) = -\cos(\mathbf{Q}(n), \bar{\mathbf{R}}(s)) \quad (7)$$

where n and s indicate token indices of $\mathbf{Q}$ and $\bar{\mathbf{R}}$, respectively, and $\cos(\cdot, \cdot)$ is a cosine similarity between two input vectors.

Then, new tokens in $\mathbf{Q}$ with higher scores than old tokens in $\mathbf{R}$ are selected and, newly pushed into the semantic repository $\mathbf{R}$. When a class holds more than K tokens, the old tokens with lower scores are discarded to maintain K representative tokens for each semantic class in $\mathbf{R}$. Updating $\mathbf{R}$ with tokens that exhibit both high classification and novelty scores enables the repository to capture rich semantic information, effectively covering dynamic driving

scenes. Note that the semantic repository $\mathbf{R}$ is updated only during the SOAP’s training phase to ensure stable inference performance.

3.4. Loss Functions

We perform the Hungarian matching [15] between predicted and ground-truth voxel masks following [4, 6, 53]. For each matched pair, we compute a mask loss L_{mask} , a mask dice loss L_{dice} , and a class loss L_{cls} as in [6]. The final loss is defined as

$$L = \lambda_{\text{mask}} L_{\text{mask}} + \lambda_{\text{dice}} L_{\text{dice}} + \lambda_{\text{cls}} L_{\text{cls}} \quad (8)$$

where, scaling weights λ_{mask} , λ_{dice} , λ_{cls} are set to 5, 5, 2, respectively.

4. Experiments

4.1. Datasets

We evaluate the performance of SOAP on two outdoor 3D SSC datasets: SemanticKITTI [1] and SSCBench-KITTI360 [21]. Both datasets provide front-view RGB images, LiDAR sweeps, and semantic scene labels represented as $256 \times 256 \times 32$ voxel grids. The scene labels cover a range of $51.2m \times 51.2m \times 6.4m$ from the front view of the vehicle.

SemanticKITTI [1]. It contains a total of 22 sequences, which are split into 10/1/11 for train/val/test following [3]. We utilize the RGB images of cam-2 with a 1280×384 resolution as model inputs. In the semantic labels, each voxel

Table 2. Semantic scene completion results on the SSCBench test set. † represents methods that use temporal information. We highlight the best results in **bold** and the second best results in underline.

Method	SC IoU	SSC mIoU	car	bicycle	motorcycle	truck	other-veh.	person	road	parking	sidewalk	other-ground	building	fence	vegetation	terrain	pole	traf.-sign	other-struct.	other-object
MonoScene [3]	37.87	12.31	19.3	0.4	0.6	8.0	2.0	0.9	48.4	11.4	28.1	3.2	32.9	3.5	26.2	16.8	6.9	5.7	4.2	3.1
TPVFormer [12]	40.22	13.64	21.6	1.1	1.4	8.1	2.6	2.4	53.0	12.0	31.1	3.8	34.8	4.8	30.1	17.5	7.5	5.9	5.5	2.7
OccFormer [50]	40.27	13.81	22.6	0.7	0.3	9.9	3.8	2.8	54.3	13.4	31.5	3.6	36.4	4.8	31.0	19.5	7.8	8.5	7.0	4.6
VoxFormer† [22]	38.76	11.91	17.8	1.2	0.9	4.6	2.1	1.6	47.0	9.7	27.2	2.9	31.2	5.0	29.0	14.7	6.5	6.9	3.8	2.4
Symphonies [13]	44.12	<u>18.58</u>	30.0	1.9	<u>5.9</u>	25.1	12.1	8.2	54.9	13.8	32.8	6.9	35.1	<u>8.6</u>	<u>38.3</u>	11.5	14.0	9.6	14.4	11.3
SGN† [26]	<u>47.06</u>	18.25	29.0	<u>3.4</u>	2.9	10.9	5.2	3.0	<u>58.1</u>	<u>15.0</u>	<u>36.4</u>	4.4	<u>42.0</u>	7.7	38.2	<u>23.2</u>	<u>16.7</u>	<u>16.4</u>	9.9	5.8
SOAP† (proposed)	48.12	20.92	<u>29.9</u>	5.6	7.8	<u>14.4</u>	<u>7.6</u>	<u>6.1</u>	60.9	17.4	40.3	<u>5.4</u>	45.3	10.6	40.5	24.8	16.8	21.0	<u>12.6</u>	<u>9.9</u>

is annotated with 21 semantic classes (19 semantics, 1 free, 1 unknown).

SSCBench-KITTI360 [21]. It is a recently introduced dataset for 3D SSC task. SSCBench-KITTI360 contains 9 long sequences, which are split into 7/1/1 for train/val/test. The input resolution of the RGB images is set to 1408×384 . The scene voxels in SSCBench-KITTI360 are annotated with 20 semantic classes (18 semantics, 1 free, 1 unknown).

4.2. Implementation Details

Network Structure. Following other comparable methods [3, 12, 44, 50, 51], we employ EfficientNetB7 as our backbone. The multi-scale number N is 4 and the spatial dimensions of the multi-scale image features are $(\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32})$ of an input image, respectively. We set a spatial voxel size $X \times Y \times Z$ to $128 \times 128 \times 16$ and channel numbers C to 192. The spatial dimensions of the multi-scale voxel features are $(1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$ of $X \times Y \times Z$, respectively. We set the number of semantic tokens to $N_q = 100$, while the depth distribution bin to $D = 112$. In the repository builder, the balancing parameter for novelty scores is set to $\alpha = 0.2$ and the number of representative tokens of the semantic repository K is set to 10. We use $P = 4$ historical frames. The scene-adaptive decoder is repeated $L = 9$ times. The output voxel is upsampled to $256 \times 256 \times 32$ through the trilinear interpolation for the evaluation.

Training. We train the proposed SOAP for 30 epochs. We employ AdamW [24] optimizer with an initial learning rate 0.0001 and divide it by 10 at the 20 and 25 epochs. For the training, we use a mini-batch of size 4 on 4 RTX A6000 GPUs. For data augmentation, we use random resize, rotation, and flip for the image space and 3D flip for the 3D volume space following [10, 20, 49].

4.3. Main Results

For evaluation on SemanticKITTI and SSCBench, we use the mean intersection over union (mIoU) of semantic

Table 3. Complexity-accuracy comparison on SemanticKITTI validation set. † represents methods that use temporal information.

Method	Image encoder	Params.	GFLoPs	IoU	mIoU
OccFormer [50]	EfficientNetB7	202.7M	973.1	36.50	13.46
HTCL† [18]	EfficientNetB7	181.4M	2622.4	45.51	17.13
SOAP†	EfficientNetB7	85.2M	705.3	47.24	19.21
Symphonies [13]	ResNet50	59.3M	611.9	41.92	14.89
HASSC† [37]	ResNet50	58.1M	313.9	44.58	14.74
SGN† [26]	ResNet50	27.9M	723.7	46.21	15.32
SOAP†	ResNet50	45.1M	627.8	48.12	18.80

classes for the SSC task and the intersection over union (IoU) for the class-agnostic scene completion (SC) task.

Highest mIoU and IoU Scores. Tables 1 and 2 compare the proposed SOAP with recent vision-centric 3D SSC algorithms on the SemanticKITTI and SSCBench-KITTI360 test sets, respectively. The scores of other methods come from respective papers. In Table 1, we observe that SOAP significantly outperforms the existing methods on both datasets in terms of the mIoU and IoU scores. Compared to HTCL [18], SOAP improves +2 SSC mIoU and +1.86 SC IoU on SemanticKITTI test set, which achieves 12% and 4% relative improvement. Also, SOAP surpasses SGN [26] on SSCBench-KITTI test set with a margin of +2.67 mIoU and +1.06 IoU. The highest mIoU and IoU scores demonstrate that the proposed SOAP faithfully refines occluded regions in the initial voxel features, while the scene-adaptive decoder further enhances the semantic context of the query embeddings.

Efficiency Evaluation. We replace our image encoder, EfficientNetB7 [35] with a more light-weight network, ResNet50 [9]. Table 3 shows complexity-accuracy comparisons between SOAP and other methods on SemanticKITTI validation set, using either ResNet50 or EfficientNetB7 as the image encoder. Compared to OccFormer [50] and HTCL [18], which utilize EfficientNetB7, SOAP requires the lowest parameters and GFLoPs while achieving the highest IoU and mIoU scores, demonstrating the effective-

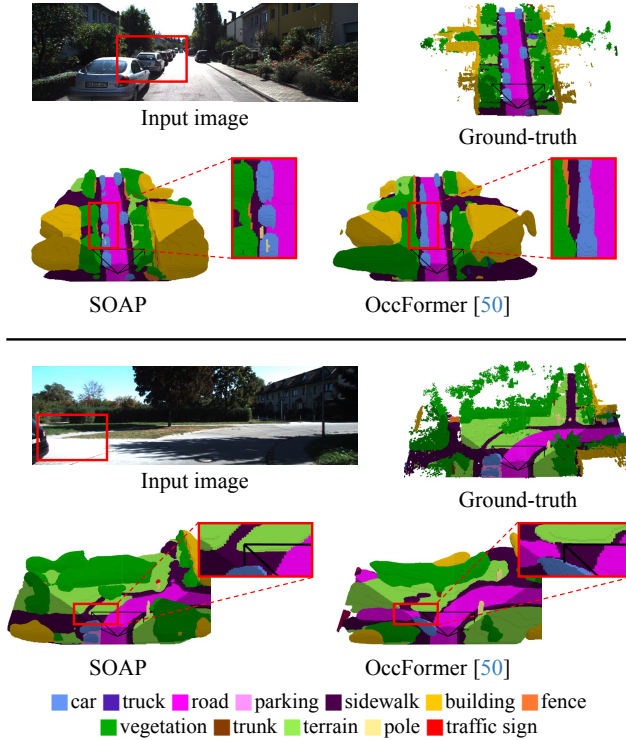


Figure 5. Qualitative results on SemanticKITTI validation set. SOAP demonstrates robustness in handling the occluded regions (top) and distinguishing similar classes (bottom). Red boxes highlight failure cases of OccFormer [50] for comparison.

ness and efficiency of SOAP. Also, SOAP with ResNet50 yields remarkable mIoU and IoU performance with comparable complexity compared to other methods.

Qualitative Results. Figure 5 presents qualitative comparisons between SOAP and OccFormer [50] in terms of occluded region reconstruction and similar class discrimination. In Figure 5 (top), OccFormer fails to recognize occluded regions behind car objects, resulting in fused car shapes, whereas SOAP appropriately distinguishes and separates the cars that are actually apart. Also, as in Figure 5 (bottom), Occformer confuses classes with similar appearance (road $\leftrightarrow$ sidewalk) while SOAP reasonably differentiates these classes.

4.4. Ablation study

We analyze the contributions of individual components of the proposed SOAP through ablation experiments on the SemanticKITTI validation set. Table 4 reports the contributions of the occluded region-aware projection (OAP) and the scene-adaptive decoder (SAD). For the results without SAD, we replace it with a standard transformer decoder [6, 50, 51]. For the results without OAP, we input the initial voxel features $\tilde{\mathbf{F}}_i^{3D}$ directly into the voxel

Table 4. Ablation study on different combinations of occluded region-aware projection (OAP) and scene-adaptive decoder (SAD).

OAP	SAD	SC IoU	SSC mIoU
		43.32	16.86
	✓	43.51	17.22
✓		46.50	18.99
✓	✓	47.24	19.21

Table 5. Alternative approaches for historical information.

Method	SC IoU	SSC mIoU
Addition	45.00	18.05
DCA	45.90	18.53
OAP	47.24	19.21

encoder, without using any historical information. In Table 4, IoU and mIoU scores show significant degradation without either OAP or SAD. This indicates that OAP effectively refines occluded regions during its voxel generation, while SAD provides representative semantic features across dynamic driving scenes for query embeddings. Notably, SOAP still retains reasonable performance even without OAP, which implements 3D SSC without temporal information.

To validate the effectiveness of OAP, we design several alternative approaches to utilize historical information. First, we simply add the historical voxel features $\mathbf{F}_i^{\text{hist}}$ to the initial voxel features $\tilde{\mathbf{F}}_i^{3D}$. Second, we employ the standard deformable cross-attention (DCA) [53] between the initial voxel features $\tilde{\mathbf{F}}_i^{3D}$ and the historical voxel features $\mathbf{F}_i^{\text{hist}}$ without occluded region separation. In Table 5, the proposed OAP achieves better IoU and mIoU scores than the standard DCA, with a margin of +1.34 and +0.68, respectively. Therefore, perception of occluded regions during view projection is essential for accurate 3D semantic scene reasoning.

5. Conclusion

We proposed a novel 3D semantic scene completion algorithm, called SOAP. The main contributions of SOAP are the occluded region-aware view projection and the scene-adaptive decoder. The occluded region-aware view projection addresses the information deficiency in vision-centric 3D SSC by refining occluded voxel regions with historical observation. The scene-adaptive decoder creates the comprehensive semantic repository using the repository builder. Then, the scene-adaptive decoder integrates informative clues from the semantic repository into query embeddings. Experiments demonstrated that SOAP significantly outperforms the state-of-the-arts on the SemanticKITTI and SSC-Bench datasets, with a margin of +2 and +2.67 mIoU scores, respectively.

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